Anthony Meza

Ph.D. Candidate

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EDUCATION

Massachusetts Institute of Technology Cambridge, MA
Ph.D. in Physical Oceanography *2021 - 2026 (Expected)*

University of California, Irvine Irvine, CA
B.S. in Mathematics - Concentration in Data Science *2018–2021*

EXPERIENCE

Woods Hole Oceanographic Institution Sep 2021–Present
Graduate Research Assistant *Woods Hole, MA*

- Used numerical ocean and climate models to investigate mechanisms controlling ocean circulation and heat transport
- Analyzed satellite products using statistical methods (i.e., multivariate, time series, and spatial analysis), to identify links between coastal sea surface temperatures and extreme California precipitation events
- Developed Python and Julia tools to process, visualize, and interpret terrabytes of climate data in high-performance computing environments

Foundation for Resilient Societies Jan 2025-Feb 2025
Technical Consultant *Cambridge, MA*

- Ran and debugged Strategic Energy & Risk Valuation Model (SERVM) simulations to assess U.S. electrical grid capacity adequacy under varying generation scenarios (e.g., solar adoption).
- Led a team of 12 undergraduate electric grid modeling interns for 4 weeks to develop an internal user guide for running SERVM experiments and interpreting model output.

Los Alamos National Laboratory Jun 2021–Aug 2021
Parallel Computing Summer Fellow *Los Alamos, NM*

- Implemented and evaluated reduced-precision in the Energy Exascale Earth System Model (E3SM), written in Fortran, to reduce computational cost and energy consumption in global climate simulations

PROJECTS

Atmospheric Physics using AI Apr 2024–Jul 2024
Kaggle Competition Participant *Remote*

- Ranked 230/694 on the public leaderboard for emulating subgrid-scale atmospheric processes
- Developed machine learning architectures with physical constraints in PyTorch and TensorFlow
- Optimized data pipelines for GPU-accelerated training, significantly increasing model throughput

xbuoy
Personal Project

- Developed *xbuoy*, a Python package for querying National Data Buoy Center observations and converting irregularly sampled buoy data to NetCDF and Zarr

SKILLS

Languages: *Programming*: Python, Julia, MATLAB; *Human*: English, Spanish
Scientific Computing: Scientific Python (xarray, PyTorch, Tensorflow) and Julia (Optimization.jl, JuMP.jl)
HPC & Dev Tools: Unix, HPC job schedulers (e.g., Slurm), Dask, Git, GitHub